SUPPLEMENTARY MATERIALS (Heart attack risk prediction)

IMPLEMENTATION DETAILS

Preprocessing

- The dataset is loaded and checked for missing values and duplicates.
- The values of each column are checked for consistency and made sure grammatical errors are corrected.
- **Summary statistics** are obtained for each class and **Correlation analysis** is performed to identify the relationship between the features and the target; the **correlation heatmap (**Appendix 1**)** revealed low correlation between the features and target (a linear kernel in SVM would struggle). Age has the highest correlation coefficient of 0.24 and diastolic blood pressure and systolic blood pressure are correlated with each other and multicollinearity may occur if both are included in the models.
- **The distribution of variables** (Figure 1), and **outlier analysis** (Appendix 2) using boxplots are performed. The target variable is slightly imbalanced and SMOTE may be considered. We also observe extreme outliers in some of the features and as such Robust Scaling will be applied during modelling.

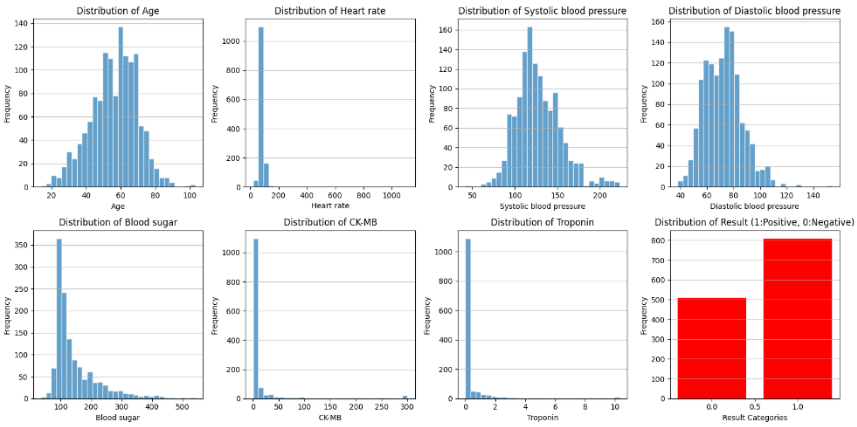


Figure 1: Distribution of Variables

- Using the difference between Systolic Blood Pressure and Diastolic Blood Pressure, a **new feature** called **Pulse Pressure** is created. It represents the force the heart generates with each beat. A low pulse pressure may indicate that the heart is not pumping enough blood which may lead to heart failure while high pulse pressure indicates future problems for people who are not physically active [1].

Modelling

- The dataset, with 1,319 observations, is split into 80% Training and 20% Testing; this ensures there is enough data for the model to learn while keeping a meaningful amount for the test set. A stratified 10-fold cross validation is implemented for a more robust and generalised models since a minor class imbalance is observed; this involves one fold for testing and nine folds for training until each fold has been used as the test set once.
- A baseline model with the following features (Table 1) is established without any hyperparameter tuning and regularisation for both SVR and MLP. Since, this is a classification task; accuracy is used as the main evaluation metric to compare the models and monitor changes when regularisation terms are introduced and during hyperparameters tuning. For the initial implementation of the MLP, 'MLPClassifier' from Scikit-learn is used but PyTorch is also used later on to investigate performance differences between the two frameworks.
- **Feature selection** (Permutation Importance) is done for better generalization.
- **Grid search** is used to find the optimal hyperparameters for both SVM and MLP. For SVM, the original hyperparameters being tuned are the regularization parameter (C), Kernel and Gamma (Kernel coefficient); and for MLP, hidden neurons/layers, activation function, backpropagation algorithm, L2 regularization, learning rate and maximum iterations (early stopping is added later on).

Table 1: Sets of features evaluated in baseline models

X1	Age, Gender, Heart rate, Systolic Blood pressure, Blood sugar, CK-Mb and Troponin
X2	Age, Gender, Heart rate, Pulse pressure, Blood sugar, CK-Mb and Troponin
Reduced Features	Age, Pulse pressure, Blood sugar, CK-Mb and Troponin

INTERMEDIATE RESULTS

Table 2: Experimental Model Results (Evaluation metric: Accuracy)

	Base SVM	**Base MLP**		**Optimised SVM**	**Optimised MLP**	
		SK learn	**PyTorch**		**SK learn**	**PyTorch**
X1	0.6607	0.7557	0.9318			
X2	0.6664	0.7586	0.9394			
Reduced features	0.7888	0.9470	0.9555	0.9593	0.9659	0.9650

Base models, Feature importance and Grid Search

The base model for SVM and SK-Learn MLP, as seen from Table 2 above, resulted in approximately 66% and 75% respectively for the two set of features the models were initially trained with. The base MLP models failed to converge with 200 maximum iterations. With Permutation Importance, we identified a reduced set of features which led to drastic improvements in performance. Random Forest was considered for feature importance but has been omitted since we wanted to identify the features important for each specific model. The set of features, X2, was used during feature importance identification since higher accuracy was observed from initial models ran. Age, CK-MB and Troponin are important factors in both models. When the models were trained later on using PyTorch, accuracy has been above 90% across the different set of features while we expected similar results to SK Learn.

For Grid search, we used the 'Reduced' set of features. SVM improved with tuned hyperparameters (79% to 96% accuracy). Grid search for MLP was ran using the Scikit Learn Classifier instead of PyTorch for easier implementation and compatibility. A fixed number of iterations of 500 did not lead to convergence. We then investigated different parameters grids. The number of hidden neurons were increased, early stopping and different number of maximum iterations was introduced. We then used less hidden neurons to monitor changes and found out that the model converges. Early stopping seemed to have been the key hyperparameter to enable convergence for sci-kit learn MLP. We used the same optimised hyperparameters obtained from sci-kit learn MLP Grid Search for PyTorch and observed a lower accuracy (which varies when the model is ran again) compared to Scikit Learn as seen in Table 2. A tailored Grid Search analysis for the PyTorch model may lead to higher accuracy but this has not been tested this time as PyTorch model is performing well already.

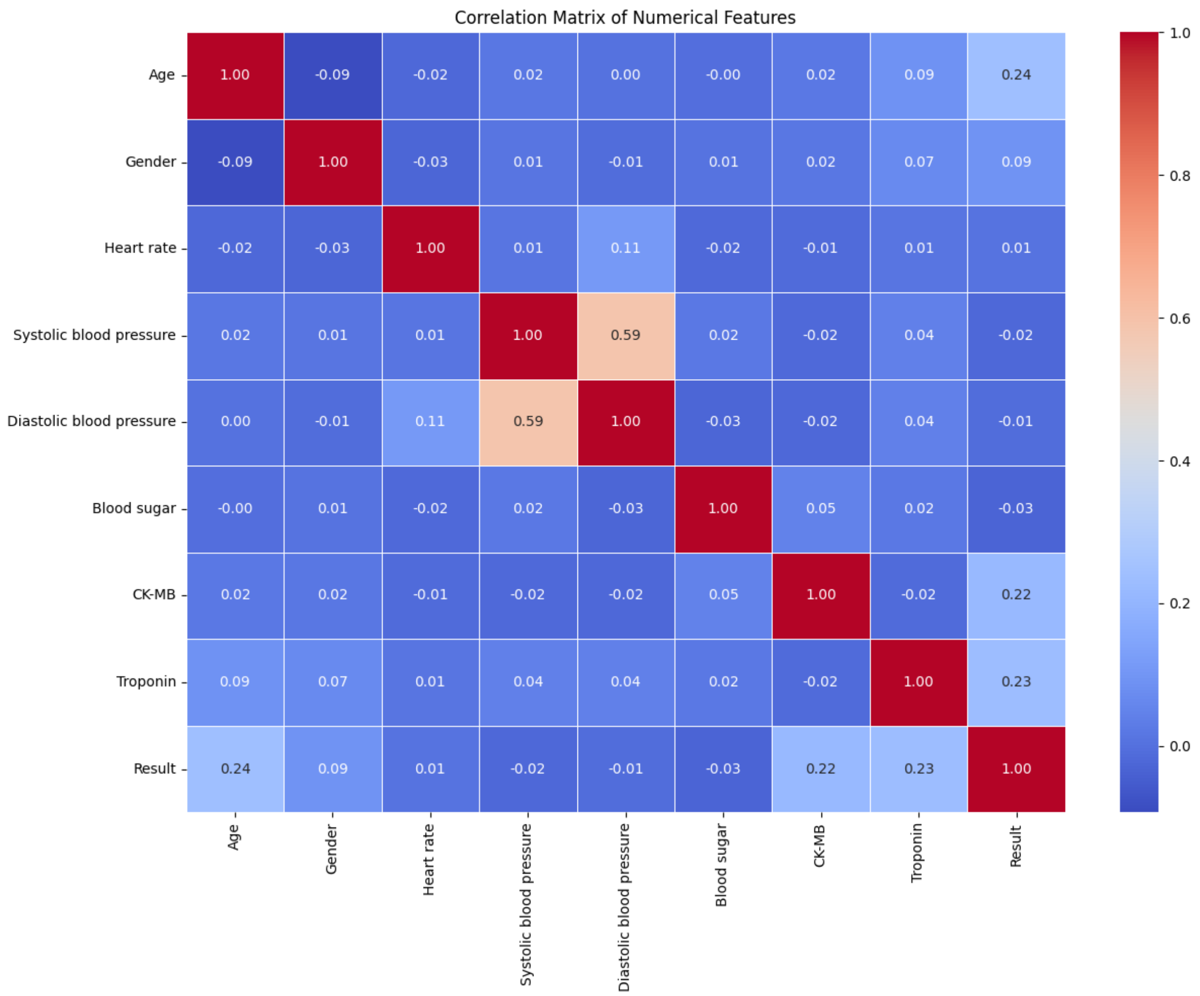
When comparing Sci-kit learn against PyTorch implementation, despite the latter being more tedious, we have more control over the layers and better results are observed. Sci-kit learn uses automatic differentiation to compute gradients and optimise weights. PyTorch implements backpropagation using gradient calculations with loss.backward() and optimizer.step() in the training loop which could have most likely led to better performance.

Glossary
Systolic Blood Pressure: The pressure when the heart contracts and pumps blood.
Diastolic Blood Pressure: The pressure when the heart relaxes between beats.
CK-MB: An enzyme released when heart muscle is damaged, used to diagnose heart-attacks.
Troponin: A protein released into bloodstream during heart damage (cardiac injury marker).
ReLU: Activation function outputting x if positive, 0 if negative, solving vanishing gradients**.**
Sigmoid: Activation function that squashes inputs to values between 0 and 1**.**
Regularization: Technique that prevents overfitting by adding penalties during training.

References

[1] Cleveland Clinic (n.d.) *Pulse pressure*. Available at: https://my.clevelandclinic.org/health/body/21629-pulse-pressure (Accessed: 22 March 2025).

Appendix 1: Correlation Heatmap



Appendix 2: Boxplot of features